

Temporal Maintenance of Vulnerability Detectors: Pre/Post-Fix and Clean-Specificity Evidence

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ABSTRACT

Learning-based vulnerability detectors are usually evaluated as static predictors, but deployed detectors are maintained software artifacts: projects, labels, thresholds, and alert-cost constraints all change over time. We introduce *MAINT-VD*, a maintenance-evaluation framework organized around chronological aging, unseen-project temporal testing, independent-clean specificity, and operating-point sweeps. The framework is instantiated on CVEfixes, recovered-date DiverseVul, and a committed PrimeVul-paired sensitivity study. On CVEfixes, a character 4-gram baseline reaches mean future recall 0.882, but repository-clean specificity is only 0.155, showing that high pre/post-fix recall is not evidence of arbitrary clean-code behavior. Under unseen-project temporal testing, a small recent-update policy reaches recall 0.972 at FPR 0.945 on CVEfixes, while the same policy reaches recall 0.347 at FPR 0.195 on recovered-date DiverseVul; no tested DiverseVul policy satisfies recall ≥ 0.50 at FPR ≤ 0.30 . Full-dataset CodeBERT and LineVul runs further show that threshold rules matter: recovered-date DiverseVul has moderate AUROC but low AUPRC, and a fixed-FPR threshold can collapse recall to nearly zero. These results show that maintenance policies and threshold rules are part of detector behavior under project and label-distribution shift; temporal recall is meaningful only when reported with false-positive and clean-specificity evidence.

1. Introduction

Learning-based vulnerability detectors are increasingly proposed as support tools for secure code review, triage, and continuous integration. Their appeal is clear: modern software projects are too large for manual inspection alone, and vulnerability reports often arrive after similar mistakes have already spread across related components, dependencies, and forks. Prior work has therefore developed graph-based detectors such as Devign [1], transformer-based detectors such as LineVul [2], and pretrained code representations such as CodeBERT [3]. Public datasets, including BigVul [4], CVEfixes [5], and DiverseVul [6] have also made vulnerability-detection experiments easier to reproduce and compare.

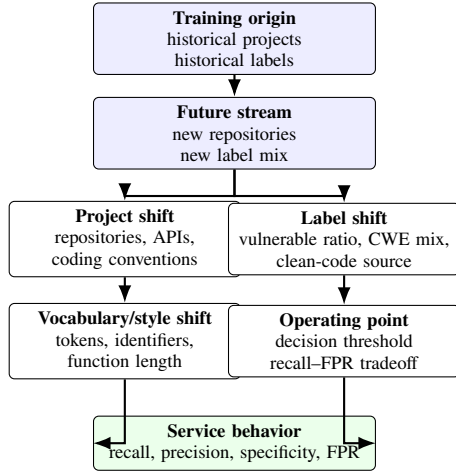
The dominant evaluation style in this literature still treats a detector as a static predictive artifact. A dataset is collected, split into training and held-out examples, and used to compare model families. This design is valuable for controlled benchmark comparison, but it omits a central property of a security detector after deployment: the detector is maintained in a changing environment. Future code is not merely additional benchmark data. Projects adopt new libraries and APIs; coding idioms change; build systems, generated code, and dependency interfaces evolve; vulnerability taxonomies and disclosure practices shift; and the projects seen after deployment may differ from those that produced the training labels. Even if model weights remain fixed, the threshold can become inappropriate as class priors and false-positive

costs change. We use *project shift* for this project-level change in the evaluation population: a detector trained on historical functions from one set of repositories may later be applied to different repositories, or to a different mixture of repositories, with different APIs, naming conventions, file structures, vulnerability types, and fix styles. We use *label-distribution shift* for changes in what the labels mean or how often they occur: for example, the future stream may contain a different vulnerable-to-clean ratio, a different mix of CWE categories, or fixed-code negatives that differ from independently clean functions. Related shift signals used in this article are descriptive measurements of that changing evaluation environment. *CWE shift* denotes changes in the weakness-category mix, *token or vocabulary shift* denotes new identifiers, APIs, or code idioms, and *length or style shift* denotes changes in function size or code structure. An *operating point* is the decision threshold at which a detector is run; changing it can trade recall against precision, specificity, and false-positive rate even when the model scores are unchanged.

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Shift terminology at a glance. Future detector behavior depends on who writes the code, what labels are observed, how code changes, and where the decision threshold is set.

This mismatch matters in practice. A detector with high recall but poor specificity can flood developers with alerts and lose trust. A detector tuned to suppress false positives may miss the vulnerabilities it was deployed to surface. A retraining policy may improve recall on one temporal split while hurting precision or failing on unseen projects. A validation set made only of fixed versions of vulnerable functions may not represent independently clean code. These are maintenance questions as much as modeling questions: when to recalibrate, how many recent labels to acquire, whether a sliding window is preferable to cumulative retraining, and which distribution shifts should trigger intervention.

We study these issues as *security detector aging*. The phrase does not imply that every detector must decay monotonically. Instead, it names the empirical problem of understanding how detector behavior changes across future temporal windows and maintenance choices. The thesis is deliberately precise: maintenance policies and threshold rules are part of detector behavior under project and label-distribution shift; temporal recall is meaningful only when reported with false-positive and clean-specificity evidence. Under this thesis, a detector is not just an architecture; it is a maintained software artifact composed of training data, feature extraction, model weights, calibration data, threshold policy, monitoring signals, and update schedule.

To make the thesis testable, we introduce *MAINT-VD* (Maintenance Evaluation for Vulnerability Detectors): a reusable empirical framework for temporal detector maintenance. The contribution is not a new vulnerability detector and not a claim that the selected model families exhaust the state of the art. Rather, MAINT-VD specifies the maintenance evaluation surface that any detector family should face: chronological splits, project-held-out temporal testing, duplicate and near-duplicate controls, independent-clean specificity checks, operating-point curves, maintenance-policy comparisons, drift-performance analysis, and monitoring-trigger calibration. CVEfixes is the primary study because the local mirror contains explicit temporal metadata; we frame it as pre/post-fix detection,

with vulnerable code versions as positive examples and fixed versions as negative examples. DiverseVul provides sensitivity evidence after enrichment with repository commit dates from the official metadata; rows whose dates cannot be recovered are excluded from recovered-date temporal claims.

The four checks are intentionally concrete. *Chronological testing* trains only on past code and evaluates on later years; for example, a detector trained on 2010–2016 CVEfixes records is tested on 2018–2024 records without seeing future labels. *Project-held-out testing* adds a deployment-style constraint: future examples from projects already seen during training are removed, so a model trained on historical functions from one project cannot receive credit for later functions from that same project. *Independent-clean specificity* asks whether a detector that separates vulnerable and fixed versions also avoids flagging clean functions collected outside vulnerability fixing commits; for example, repository-clean C/C++ functions are used to measure how often a CVEfixes-trained model raises false alarms on code that is not constructed as a fixed-code negative. *Operating-point sweeps* report recall, precision, F1, specificity, and false-positive rate across thresholds rather than at one F1-optimized threshold; for example, a threshold that maximizes recall may be contrasted with one constrained to validation FPR ≤ 0.20 .

The study is guided by four research questions:

- RQ1:** How do fixed-origin vulnerability detectors behave when evaluated on later temporal windows?
- RQ2:** Which measurable distribution shifts are associated with temporal performance changes?
- RQ3:** Which maintenance policies recover recall or F1 most effectively, and at what labeling cost?
- RQ4:** What can a small-window monitoring calibration reveal once service-level maintenance events are defined?

The article makes three contributions:

1. *MAINT-VD*¹, a focused maintenance-evaluation framework organized around four reusable checks: chronological testing, project-held-out testing, independent-clean specificity, and operating-point sweeps.
2. Three instantiated evaluation cases with reusable validity controls: a corrected temporal CVEfixes pre/post-fix study, a recovered-date DiverseVul sensitivity study, and a true-dated PrimeVul-paired sensitivity study, each supported by duplicate, grouping, project-held-out, and independent-clean checks.
3. Detector-family demonstrations, not a detector leaderboard: lightweight classical baselines, frozen CodeBERT embeddings, calibrated LineVul, and

¹<https://github.com/LeoTTTPhat/MAINT-VD>

Table 1
Findings summary.

Finding	Empirical support
Clean specificity changes the interpretation of recall	The CVEfixes character model has mean future recall 0.882 in ordinary temporal testing, but temporal FPR is high and repository-clean specificity is 0.155.
Maintenance policies are operating-point choices	Under unseen-project CVEfixes testing, small recent update reaches recall 0.972 with FPR 0.945; the gain is meaningful only if that alert rate is acceptable.
DiverseVul C2 is a negative service-level result	No recovered-date DiverseVul policy reaches recall at least 0.50 while keeping FPR at or below 0.30. Among FPR-compliant policies, CWE-aware sliding has the highest recall, 0.430 at FPR 0.290; small recent update is more conservative, 0.347 at FPR 0.195.
Recovered-date DiverseVul is sensitivity evidence	The recovered-date subset supports temporal analysis but narrows coverage to 37,425 post-deduplication rows and 156 projects.
Neural zero-recall results are not a positive finding	Recovered-date DiverseVul neural rows have validation recall 0.000 at the selected FPR-controlled threshold, so they are treated primarily as a threshold-rule artifact.
PrimeVul-paired is now commit-dated but still paired	The PrimeVul-paired check recovers commit dates for 494 commit pages and keeps 1,267 post-deduplication rows; it remains a pre/post-fix task rather than arbitrary clean-code evaluation.

full-dataset CodeBERT/LineVul fine-tuning are used to exercise MAINT-VD's maintenance checks under explicit false-positive and threshold policies.

The remainder of the article is structured as follows. Section 2 reviews vulnerability detection, dataset-validity threats, temporal evaluation, and concept drift as software-maintenance concerns. Section 3 presents the MAINT-VD protocol. Section 4 defines the experimental design, including datasets, temporal splits, detector families, maintenance policies, validity controls, and the statistical protocol. Section 5 reports the empirical results for temporal behavior, policy tradeoffs, neural operating points, clean-specificity checks, drift-performance links, and monitoring calibration. Section 6 discusses practical implications, Section 7 summarizes threats to validity, and Section 8 concludes with the maintenance lesson for future vulnerability-detector evaluations.

2. Background and Related Work

2.1. Vulnerability Detectors as Maintained Tools

Software engineering tools require maintenance. Static-analysis rules become obsolete, dependency scanners need updated advisories, test suites require new fixtures, and CI policies are recalibrated as projects evolve. Learning-based vulnerability detectors have the same character. The model, feature extractor, training data, calibration threshold, and monitoring policy are all maintenance surfaces. This view is consistent with broader software-engineering work on machine-learning systems, which treats data dependencies, configuration, thresholds, monitoring, and changes in the external world as sources of maintenance cost rather than as peripheral implementation details [7, 8].

This matters because vulnerability detection is recall-sensitive. Missed positives may allow vulnerable code to

pass review, while excessive false positives create alert fatigue and reduce adoption. A useful deployment policy therefore requires more than a single aggregate test score. It requires answers to maintenance questions: when should the detector be recalibrated, how many recent examples are needed, whether retraining helps, and which drift signals should trigger intervention.

2.2. Learning-Based Vulnerability Detection

Learning-based vulnerability detection has moved from hand-engineered features and graph representations toward pretrained code models. Devign models program semantics with graph neural networks [1]; LineVul adapts transformers to line-level vulnerability prediction [2]; CodeBERT provides pretrained code representations that are widely reused for downstream code intelligence tasks [3]. Datasets such as Big-Vul, CVEfixes, and DiverseVul have made these studies easier to reproduce by linking code snippets, vulnerability reports, and fixing commits [4, 5, 6]. However, several studies caution that high benchmark scores can be fragile when datasets contain label noise, project overlap, duplicated or near-duplicated code, unrealistic negative examples, or split designs that do not match future use [9, 10, 11, 12, 13, 14, 6]. We therefore treat detector architecture as only one part of the problem. The empirical unit of interest is the maintained detector, including data, split design, threshold, and update policy.

Recent code-model and dataset work broadens this landscape. CodeT5 and UniXcoder introduce stronger pretrained code representations than the CodeBERT/LineVul family used in our executable artifact [15, 16]. PrimeVul emphasizes benchmark hygiene and FPR-aware evaluation, which is closely aligned with our operating-point argument [17]. ReposVul and ICVul move dataset construction toward repository-level evidence, patch untangling, and richer vulnerability metadata, including vulnerability-contributing commits [18, 19]. These corpora

are promising targets for future MAINT-VD replications. We do not claim architectural completeness; instead, we use classical, frozen-embedding, CodeBERT, and LineVul runs to test whether maintenance conclusions survive across low-cost and neural detector families.

Dataset construction is especially important for vulnerability detection because labels are often inferred from vulnerability-fixing commits. This practice creates several risks. A fixing commit can contain refactoring, tests, formatting changes, or defensive changes that are not themselves vulnerable. Functions touched by the same commit can appear in both vulnerable and non-vulnerable roles, and near-duplicate code can cross random or temporal split boundaries. Negative examples constructed from fixed versions of vulnerable functions are useful for a pre/post-fix task, but they do not prove that a detector distinguishes vulnerable code from independently clean code. These dataset-bias mechanisms motivate the article's explicit checks for exact duplicates, near duplicates, project and commit grouping, and independently sampled clean functions.

2.3. Temporal and Project Generalization

Random train/test splits can overestimate performance when examples from the same projects, vulnerability families, or near-duplicate patches appear across splits. DiverseVul explicitly reports performance drops under unseen-project evaluation and discusses label-noise risks in vulnerability-fixing commit datasets [6]. These concerns are not only benchmark concerns. They mirror deployment: a security tool trained on historical data must operate on future code, often in projects that were not represented in the training window.

Temporal evaluation is also related to concept drift. In streaming and continual-learning settings, concept drift refers to changes in the joint distribution of features and labels over time, and practical responses include monitoring, recalibration, windowed retraining, and active collection of new labels [20]. Vulnerability detection adds software-engineering specific drift mechanisms: dependency ecosystems change, generated code patterns appear, projects adopt new APIs, and public vulnerability reporting practice changes. These mechanisms motivate measuring project turnover, CWE mix, token distribution, code length, class prior, and clean-code specificity rather than reporting a single aggregate future score.

Time-aware evaluation also appears in perspective-testing protocols that train only on information available before a cutoff and test on later observations to avoid the benefit of hindsight. MAINT-VD shares that constraint but adds a maintenance layer: after constructing chronological and unseen-project splits, it asks how threshold selection, calibration, retraining, label budget, and clean-code specificity change service behavior. The contribution is therefore not another random-split detector benchmark, but an evaluation pattern for detectors that must be operated and maintained after initial training.

2.4. Positioning of This Study

The article differs from architecture-centered detector articles in three ways. First, it evaluates detector behavior over chronological windows and unseen-project splits rather than only random held-out examples. Second, it compares maintenance actions: no refresh, cumulative retraining, sliding-window retraining, calibration-only updates, and small recent updates. Third, it reports operating points and false-positive costs explicitly. The intended contribution is therefore methodological and maintenance-oriented: it shows how to evaluate whether a vulnerability detector remains useful as its software environment changes.

3. MAINT-VD Protocol

MAINT-VD is best understood as a four-component maintenance evaluation: chronological testing, project-held-out testing, independent-clean specificity, and operating-point sweeps. The framework is detector-agnostic: a future study can replace the classical or neural models used here while keeping the same maintenance checks, split logic, cost accounting, and artifact manifest. The implementation below gives the operational steps used to make those checks reproducible. Most steps instantiate standard empirical-SE practice, including dataset checks, chronological splits, baseline construction, statistical summary, and artifact release. The vulnerability-detector-specific additions are the explicit drift-source analysis and monitoring-trigger calibration, which connect distribution shift to maintenance actions.

1. **Dataset feasibility.** Verify code text, labels, dates, project identifiers, and weakness metadata.
2. **Temporal splitting.** Build fixed-origin and rolling-origin splits using only past data for training and calibration.
3. **Baseline construction.** Train detector families with distinct feature assumptions.
4. **Aging-curve measurement.** Measure future recall, F1, precision, specificity, false-negative rate, and MCC by year.
5. **Drift-source analysis.** Quantify token, project, CWE, length, and class-prior shifts.
6. **Maintenance-policy evaluation.** Compare no refresh, cumulative retraining, sliding-window retraining, threshold calibration, and small recent updates.
7. **Monitoring-trigger design.** Evaluate drift signals against pre-specified service-level events.
8. **Statistical summary.** Report confidence intervals, permutation tests, and sensitivity across detector families.
9. **Artifact release.** Publish scripts, manifests, splits, metrics, and limitations.

Figure 1 gives the same idea as a deployment loop: historical training data and future projects are connected through measured drift, a maintenance policy, an operating point, and an explicit service outcome.

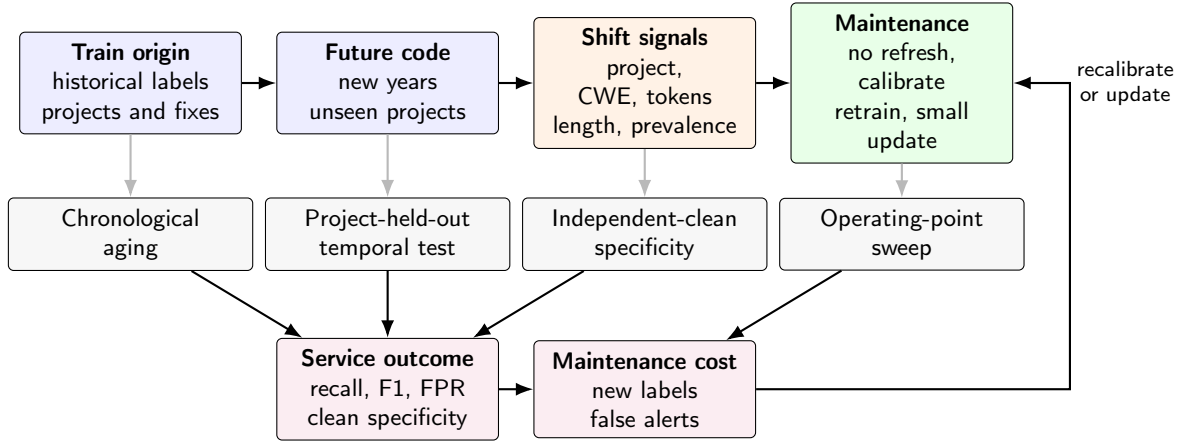


Figure 1: Visual summary of MAINT-VD detector maintenance.

4. Experimental Design

4.1. Datasets

Table 2 summarizes the temporal exports. CVEfixes is the primary source because the local mirror contains explicit temporal metadata. The normalizer converts vulnerable_code into positive records and fixed_code into negative records. This construction supports temporal maintenance analysis of a pre/post-fix detector, but it should not be read as arbitrary vulnerable-versus-clean classification.

DiverseVul is included as recovered-date sensitivity evidence. The available local mirror does not expose explicit commit dates, so using it directly would force a coarse CVE-year proxy and skip many raw rows. We therefore join the mirror to the official DiverseVul metadata file, which contains repository and commit URLs. The join matches 45,402 local rows. A broader HTTP recovery pass over GitHub commit pages and known mirrors recovers commit dates for 1,008 project/commit pairs, covering 37,767 rows before deduplication. After truncation and exact-duplicate removal, the recovered-date subset contains 37,425 rows. Rows without recovered dates are excluded from recovered-date temporal claims rather than mixed with proxy dates. The recovered-date filter also narrows project diversity: the sensitivity subset contains 156 projects, so DiverseVul is used as recovered-date sensitivity evidence rather than as a claim of broad project coverage.

PrimeVul-paired is added as a third external sensitivity dataset after commit date recovery. The public paired mirror includes commit identifiers and commit URLs but no explicit timestamp column. We therefore fetch commit pages and extract commit timestamps, recovering dates for 494 commit pages. This yields 1,393 timestamped rows before exact deduplication and 1,267 rows after deduplication. The evidence is stronger than the earlier CVE-year proxy check because the temporal split is commit-dated, but the task is still paired pre/post-fix detection and is not treated as independent clean-code classification.

4.2. Temporal Splits

The corrected CVEfixes run trains on 2010–2016, validates on 2017, and tests on 2018–2024. The recovered-date DiverseVul run trains on 2009–2015, validates on 2016, and tests on 2017–2022. Both runs require minimum positive and negative counts in the training, validation, and test windows.

For sensitivity, we rerun CVEfixes over four cutoffs: 2009–2015/2016/2017–2024, 2010–2016/2017/2018–2024, 2011–2017/2018/2019–2024, and 2012–2018/2019/2020–2024. This tests whether the CVEfixes stability claim depends on a single temporal origin. The rolling-origin drift analysis uses the same four CVEfixes origins and the analogous recovered-date DiverseVul origins: 2009–2014/2015/2016–2022, 2009–2015/2016/2017–2022, 2009–2016/2017/2018–2022, and 2009–2017/2018/2019–2022, yielding 22 DiverseVul future-year pairs.

The temporal design has two variants. The ordinary temporal variant evaluates all examples in the future year after exact duplicate removal. This measures future-window behavior under the same broad project population. The project-held-out temporal variant removes validation and test examples whose project appeared in the relevant training window. For no-refresh and calibration-only policies, the exclusion set is the original training-period project set. For cumulative, sliding-window, and small-update policies, the exclusion set is the full set of projects eligible for the policy’s training or update data; any sampling used to bound model-fitting cost does not weaken the project-holdout definition.

4.3. Detector Families

The lightweight baselines are deterministic and inspectable:

- token_tfidf_logreg: token TF-IDF features with logistic regression.
- char4_hash_logreg: hashed character 4-gram features with logistic regression.

Table 2

Temporal exports after exact duplicate removal.

Dataset	Years	Records	Removed duplicates	Vulnerable	Non-vulnerable	Projects
CVEfixes	1999–2024	19,791	2,269	8,976	10,815	3,836
DiverseVul recovered dates	2005–2022	37,425	342	2,378	35,047	156
PrimeVul-paired recovered dates	2009–2022	1,267	126	627	640	74

Table 3

DiverseVul recovered-date accounting.

Pipeline stage	Rows	Projects or pairs	Notes
Local normalized DiverseVul mirror	48,245	223 projects	Before date recovery
Matched to official metadata	45,402	1,193 project/commit pairs	Repository and commit URLs available
Recovered commit dates	37,767	1,008 project/commit pairs	10,478 rows still lack recovered dates
After truncation and exact deduplication	37,425	156 projects	342 exact duplicates removed
Fixed-origin temporal span, 2009–2022	37,143	156 projects	Train/validation/test years used below

- `code_metrics_logreg`: source-code metrics and security-token counts with logistic regression.
- `frozen_hash_embedding_logreg`: deterministic frozen random-indexing features with a trained logistic head.

We designate `char4_hash_logreg` as the primary low-compute baseline because it is language-agnostic and robust to tokenization differences. This is a protocol choice recorded in the released analysis scripts, not an external OSF-style preregistration. To address detector-family validity, we also evaluate frozen CodeBERT embeddings with a logistic head, a pretrained LineVul-style classifier calibrated on the validation year, and full-dataset fine-tuning of CodeBERT and LineVul.

Exploratory bounded fine-tuning runs update all transformer layers but use only 400 balanced training rows, 300 validation rows, 150 test rows per future year, one epoch, and maximum sequence length 128. They are retained in the artifact as execution checks and are not used for comparative claims. The stricter full-dataset neural setting uses all temporal training rows, filters validation and future test to projects unseen during training years, and selects thresholds on validation subject to a false-positive-rate ceiling of 0.20 when feasible. We therefore interpret neural results through operating-point curves rather than a single selected threshold.

4.4. Maintenance Policies and Operating Points

The maintenance-policy experiment treats a detector as a software artifact that can be refreshed in different ways after deployment. The no-refresh policy keeps the original model and threshold. Cumulative retraining uses all available years up to two years before the target test year, then calibrates on the immediately preceding year. Sliding-window retraining uses the three most recent eligible years before calibration. Calibration-only keeps the original model weights but reselects the decision threshold on recent validation data. Small recent update appends a bounded number of recent positive

and negative examples to the original training set before refitting. We add a vulnerability-specific policy, CWE-aware sliding retraining, that starts from the three-year sliding window and oversamples weakness classes whose positive share has increased in the most recent labeled calibration year. The policy is not allowed to inspect the target test year; it uses the recent labeled year to adapt the update batch toward shifted CWE prevalence.

Operating points are selected on validation data only. For classical aging and maintenance summaries, the default threshold maximizes validation F1. For the full-dataset neural project-held-out experiment, the threshold is selected subject to a validation false-positive-rate ceiling of 0.20 when feasible; if the ceiling is infeasible, the lowest-FPR candidate is reported. To avoid hiding this design choice, neural results also include threshold sweeps with precision, recall, F1, specificity, FPR, AUROC, and AUPRC. The article treats recall gains without FPR context as incomplete maintenance evidence. In the reported full-dataset neural runs, the FPR ceiling is feasible for all four validation selections. The recovered-date DiverseVul neural rows have validation FPR 0.000 because the selected threshold predicts no positives on validation, not because the ceiling is infeasible. For this imbalanced project-held-out setting, the fixed-FPR rule is therefore treated as mis-specified: it can satisfy the service ceiling by suppressing nearly all positive predictions. A proper neural rerun should compare this rule with precision-floor, best-F1, and Youden-*J* threshold policies and report the full validation precision-recall surface.

4.5. Leakage and Validity Controls

Exact canonical duplicate functions are removed before splitting. We also run a near-duplicate control that preserves the training origin and removes validation or test records whose SimHash is within Hamming distance 3 of an earlier kept record and whose token-count ratio is between 0.75 and 1.33. The recovered-date DiverseVul validity pass finds

10,623 near-duplicate candidate pairs after exact deduplication, while grouped analysis reports no train-to-test commit-group or pair-group overlap.

To test whether fixed-code negatives substitute for independently clean code, we construct a repository-clean benchmark from local research software and external C/C++ repositories outside the vulnerability-dataset construction pipeline. The extractor keeps Python and C-like functions, removes exact canonical overlap with CVEfixes, recovered-date DiverseVul, and PrimeVul-paired, and caps per-project contribution. The resulting benchmark contains 2,500 functions from 39 project/source groups: 2,265 C/C++-family functions and 235 Python functions. These projects are not claimed to be formally defect-free; the point is narrower and stronger than the earlier dataset-derived stress test: the clean examples are not derived from vulnerability-fixing commits and do not reuse the temporal vulnerability datasets' negative-example construction.

4.6. Statistical Protocol

The statistical protocol is descriptive because the number of annual future windows is small. We therefore focus on effect sizes and uncertainty intervals rather than binary significance claims. For fixed-origin aging, the primary effect is the change between the first and last future windows and the mean future score over all test years. For maintenance policies, the primary effect is the paired difference from no refresh in each future year, reported with mean recall, mean F1, mean FPR, and mean label cost per window.

Uncertainty intervals for policy summaries are non-parametric bootstrap intervals over yearly windows using 1,000 resamples and the 2.5th and 97.5th percentiles. Drift-performance links are evaluated with rolling-origin paired replicates rather than only one fixed origin. We reuse the four CVEfixes cutoff origins from the temporal sensitivity analysis and add the analogous four recovered-date DiverseVul origins listed in Section 4.2. Each future test year under each origin contributes one paired row containing drift signals and recall drop from that origin's first future window, yielding 26 CVEfixes pairs and 22 DiverseVul pairs. Pearson correlations are checked with permutation tests that shuffle the paired performance sequence while preserving the drift series. For each dataset, the nine drift-signal tests are adjusted with the Holm procedure, and correlation intervals are reported with Fisher's z-transform. The tests are interpreted as associations, not causality. For neural operating-point analysis, AUROC and AUPRC summarize ranking quality, while threshold-specific precision, recall, F1, specificity, and FPR summarize deployment behavior at candidate operating points.

To keep the claims falsifiable, we also state two operating-point decision criteria. C1 asks whether small recent updates increase recall by at least 0.05 over no refresh on CVEfixes under unseen-project temporal testing while keeping future FPR at or below 0.20. C2 asks whether any recovered-date DiverseVul maintenance policy maintains recall at least 0.50 with future FPR at or below 0.30 across

future windows. The current results do not pass C1 because the CVEfixes recall gain is coupled to FPR 0.945, and they do not identify a DiverseVul policy that satisfies the C2 service level. This framing turns the maintenance thesis into explicit service-level tests rather than a general claim that all update policies help.

5. Results

Table 1 provides the summary anchor for the results. The subsections below give the detailed evidence behind those findings: fixed-origin temporal behavior and cutoff sensitivity, project-held-out policy tradeoffs, repository-clean specificity, neural operating-point diagnostics, rolling-origin drift correlations, and ordinary-temporal maintenance summaries.

5.1. RQ1: Fixed-Origin Temporal Behavior

Table 4 reports fixed-origin aging summaries. CVEfixes does not show catastrophic aging for the primary character baseline under the corrected split: mean future recall is 0.882 [0.871, 0.893], but temporal FPR is 0.821 and independent-clean specificity is 0.155 [0.142, 0.170]. These are pre/post-fix detection scores and are not interpreted as arbitrary vulnerable-versus-clean deployment performance. The recovered-date DiverseVul subset is more volatile: the same baseline has mean future recall 0.455 [0.371, 0.556] with temporal FPR 0.252 and independent-clean specificity 0.700 [0.682, 0.718]. This contrast is central to the article's claim. Temporal behavior is dataset- and policy-dependent; the maintenance problem is to measure that behavior and choose an operating policy, not to assume universal monotonic decay.

Table 5 reports the CVEfixes cutoff sensitivity for the primary character baseline. The ordinary-temporal stability claim is not an artifact of a single origin: mean recall ranges from 0.884 to 0.978 across four train/validation/test origins. The sign of first-to-last recall change varies, which supports the article's narrower claim that temporal behavior is origin- and policy-dependent rather than monotonically aging. In Table 5, decay is first-test recall minus last-test recall, so negative values indicate higher recall in the last test window.

Table 6 adds PrimeVul-paired as a third external sensitivity dataset with recovered commit dates. The commit-date subset is much smaller than the full paired mirror because only 494 commit pages were recovered, but its temporal split no longer depends on CVE-year proxy dates. The task remains paired pre/post-fix detection rather than arbitrary vulnerable-versus-clean detection. We therefore interpret the result as external temporal sensitivity evidence: classical detectors again show that recall on paired vulnerable/fixed examples must be interpreted jointly with alert cost and independent-clean specificity.

Table 7 adds a compact cross-dataset probe for the primary character model. Each row trains and calibrates on one dataset's historical window, then tests on future windows from another recovered-date dataset. The results should be read as a stress test of dataset bias, not as a new primary

Table 4

Headline temporal and independent-clean endpoints.

Dataset	Model	Mean recall [95% CI]	Mean F1	Temporal FPR	Clean specificity [95% CI]	Clean FPR
CVEfixes	char4 hash LR	0.882 [0.871, 0.893]	0.619	0.821	0.155 [0.142, 0.170]	0.845
CVEfixes	code metrics LR	0.904 [0.892, 0.917]	0.633	0.804	0.075 [0.065, 0.086]	0.925
CVEfixes	frozen emb. LR	0.820 [0.808, 0.831]	0.628	0.669	0.275 [0.258, 0.293]	0.725
CVEfixes	token TF-IDF LR	0.851 [0.833, 0.868]	0.653	0.637	0.184 [0.170, 0.200]	0.816
DiverseVul recovered	char4 hash LR	0.455 [0.371, 0.556]	0.180	0.252	0.700 [0.682, 0.718]	0.300
DiverseVul recovered	code metrics LR	0.655 [0.607, 0.702]	0.215	0.307	0.825 [0.810, 0.840]	0.175
DiverseVul recovered	frozen emb. LR	0.345 [0.323, 0.368]	0.180	0.170	0.812 [0.796, 0.826]	0.188
DiverseVul recovered	token TF-IDF LR	0.633 [0.599, 0.666]	0.172	0.396	0.579 [0.560, 0.598]	0.421

Table 5

CVEfixes cutoff sensitivity for the primary character baseline.

Cutoff	First recall	Last recall	Decay	Worst recall	Mean recall
2009–2015/2016/2017–2024	0.892	0.877	0.014	0.872	0.884
2010–2016/2017/2018–2024	0.866	0.902	-0.035	0.866	0.884
2011–2017/2018/2019–2024	0.951	0.977	-0.027	0.951	0.965
2012–2018/2019/2020–2024	0.976	0.985	-0.009	0.971	0.978

benchmark. Cross-dataset recall is possible, but FPR and F1 vary substantially; this reinforces the article’s claim that dataset construction and operating point are part of detector maintenance.

Table 8 repeats the five standard classical policies and the CWE-aware sliding policy under unseen-project temporal testing. This design is the strongest classical policy-validity check because it asks whether maintenance helps on projects not seen during training. On CVEfixes, refresh and calibration policies increase recall over no refresh, but mean FPR remains between 0.932 and 0.945. The recall gain therefore buys more detected positives at substantial

alert cost. The CWE-aware policy slightly lowers CVEfixes FPR relative to the other refresh policies, but it does not solve the alert-cost problem. On recovered-date DiverseVul, refresh policies reduce FPR while also reducing recall. The C2 criterion is therefore a negative result: no tested policy reaches recall ≥ 0.50 while keeping FPR ≤ 0.30 . Across both datasets, maintenance policy, label cost, and operating point must be interpreted together.

Table 9 isolates the small-update label budget because reviewers reasonably asked whether the pre-specified 40-label setting is an arbitrary sweet spot. It is deliberately modest: each future window receives at most 20 newly labeled

Table 6

PrimeVul-paired recovered-date fixed-origin sensitivity.

Model	First recall	Last recall	Mean recall [95% CI]	Mean F1 [95% CI]	Mean FPR [95% CI]
char4 hash LR	0.726	1.000	0.789 [0.675, 0.910]	0.625 [0.585, 0.666]	0.776 [0.679, 0.891]
code metrics LR	1.000	1.000	1.000 [1.000, 1.000]	0.685 [0.660, 0.716]	0.998 [0.995, 1.000]
frozen emb. LR	0.938	1.000	0.939 [0.905, 0.973]	0.671 [0.653, 0.688]	0.932 [0.883, 0.972]
token TF-IDF LR	0.779	0.900	0.810 [0.758, 0.866]	0.639 [0.617, 0.666]	0.779 [0.662, 0.906]

Table 7

Cross-dataset temporal probe for the primary character model.

Train/calibration source	Future test source	Windows	Mean recall	Mean F1	Mean FPR
CVEfixes	DiverseVul recovered	6	0.883	0.130	0.861
CVEfixes	PrimeVul recovered	5	0.912	0.653	0.892
PrimeVul recovered	DiverseVul recovered	6	0.821	0.135	0.769
DiverseVul recovered	PrimeVul recovered	5	0.322	0.394	0.300

Table 8

Classical maintenance policies under unseen-project temporal testing.

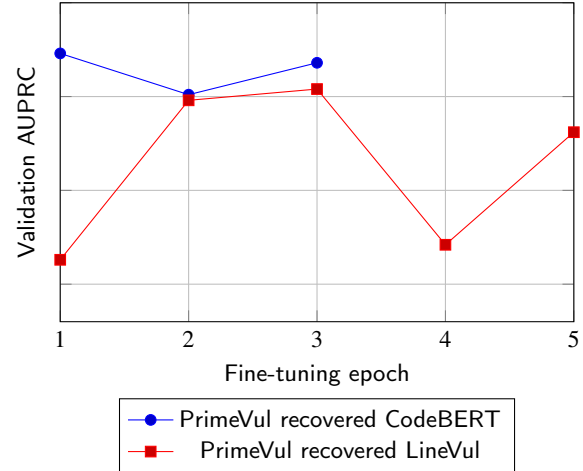
Dataset	Policy	Mean recall	Recall CI	Mean F1	Mean FPR	Labels/window
CVEfixes	No refresh	0.935	[0.925, 0.944]	0.629	0.890	0.0
CVEfixes	Cumulative retrain	0.970	[0.953, 0.985]	0.638	0.938	975.9
CVEfixes	Sliding 3-year retrain	0.971	[0.957, 0.983]	0.638	0.944	1021.0
CVEfixes	Calibration only	0.972	[0.951, 0.988]	0.633	0.944	1529.3
CVEfixes	Small recent update	0.972	[0.948, 0.989]	0.637	0.945	40.0
CVEfixes	CWE-aware sliding	0.966	[0.953, 0.978]	0.638	0.932	1021.0
DiverseVul recovered	No refresh	0.943	[0.893, 0.976]	0.149	0.860	0.0
DiverseVul recovered	Cumulative retrain	0.424	[0.196, 0.675]	0.158	0.291	1546.8
DiverseVul recovered	Sliding 3-year retrain	0.379	[0.173, 0.632]	0.162	0.243	1573.8
DiverseVul recovered	Calibration only	0.435	[0.219, 0.700]	0.152	0.315	2397.7
DiverseVul recovered	Small recent update	0.347	[0.164, 0.522]	0.174	0.195	40.0
DiverseVul recovered	CWE-aware sliding	0.430	[0.276, 0.655]	0.175	0.290	1573.8

positives and 20 newly labeled negatives, approximating a lightweight periodic triage batch rather than full retraining. The sensitivity run shows that budgets from 10 to 200 labels/window barely change the service profile. The method preserves high recall, but high FPR remains; in this setting the limiting factor is the operating point and project shift, not the exact small-update budget.

Table 10 and Figure 2 report the tuned neural diagnostic rerun requested by the review analysis. The rerun uses full-parameter fine-tuning, the recovered-date PrimeVul subset, all eligible training, validation, and test rows under future-project holdout, maximum sequence length 256, class-balanced loss, five requested epochs, early stopping by validation AUPRC, and retained validation-threshold curves. It is still a diagnostic rather than a definitive architecture comparison, but it is no longer a tiny capped run: both models train on all 499 eligible historical rows and are tested on all 329 eligible future unseen-project rows. The curves show limited learnability rather than a silent pipeline failure. CodeBERT reaches moderate validation AUPRC but the selected FPR-controlled threshold predicts almost all future examples negative; LineVul reaches a nonzero validation operating point but pays higher future FPR.

5.2. Near-Duplicate-Controlled and Neural Reruns

Table 11 reports the near-duplicate-controlled rerun and lightweight neural-transfer sensitivities. The control does not reverse the main interpretation: CVEfixes remains largely stable for the primary character model, while

**Figure 2:** PrimeVul-paired neural validation AUPRC by epoch.

DiverseVul continues to show substantial future recall decay. Frozen CodeBERT embeddings are competitive on CVEfixes; the calibration-only LineVul transfer baseline is weak. Bounded full fine-tuning is excluded from the main comparison because the sampled one-epoch, length-128 setting is too small to support a meaningful architecture claim.

Table 12 reports the stricter neural experiment. Unlike the bounded run, this design uses all training records, evaluates only future projects absent from the training period, and caps validation FPR at 0.20 when selecting thresholds.

Table 9

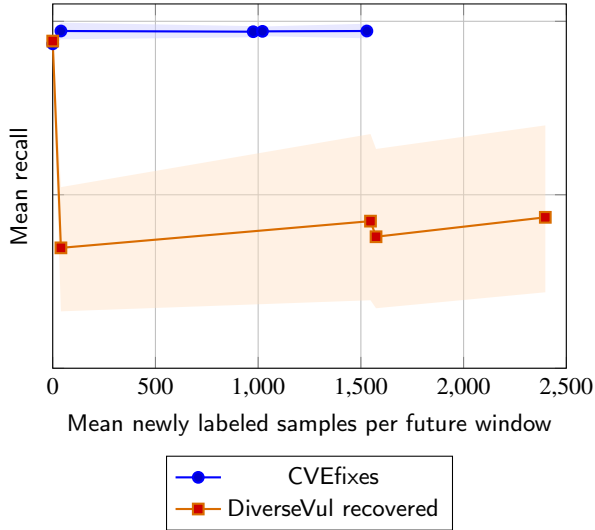
CVEfixes project-held-out small-update budget sensitivity.

Labels/window	Recall	Mean F1	Mean FPR	Interpretation
10	0.973 [0.952, 0.988]	0.637	0.947	High recall, high alert cost
40	0.972 [0.948, 0.989]	0.637	0.945	Pre-specified budget
100	0.973 [0.949, 0.991]	0.637	0.946	No material gain
200	0.966 [0.947, 0.986]	0.636	0.941	Slightly lower FPR

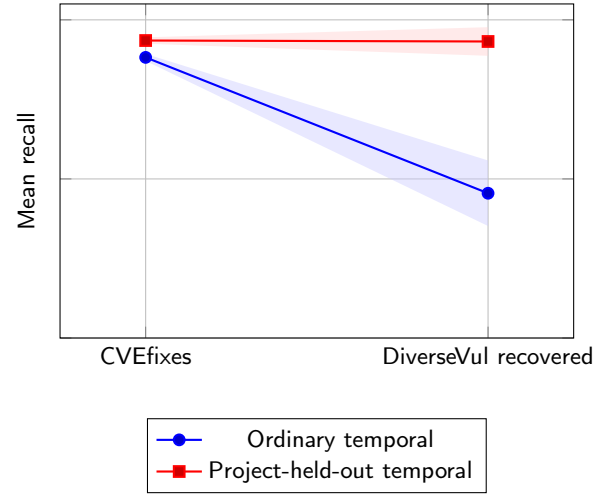
Table 10

PrimeVul-paired recovered-date neural diagnostic rerun.

Model	Sample and epoch	Validation point	operating	Future test behavior
CodeBERT full FT	Train/val/test 499/57/329; epoch 1	best	AUPRC 0.573; selected recall/FPR 0.000/0.000	Recall 0.015 [0.000, 0.037]; F1 0.027 [0.000, 0.066]; FPR 0.014 [0.000, 0.034]
LineVul full FT	Train/val/test 499/57/329; epoch 3	best	AUPRC 0.554; selected recall/FPR 0.250/0.172	Recall 0.234 [0.084, 0.376]; F1 0.280 [0.134, 0.416]; FPR 0.227 [0.085, 0.376]

**Figure 3:** Policy-vs-label-cost view under unseen-project temporal testing.

On CVEfixes, CodeBERT retains moderate recall and F1, while LineVul is more conservative. On recovered-date DiverseVul, the same FPR-controlled thresholding rule drives both neural models to very low recall. We therefore do

**Figure 4:** Ordinary temporal versus project-held-out no-refresh evaluation.

not interpret the DiverseVul neural rows as evidence that one architecture is better than another, nor as evidence that the representation fails to rank examples. They are a diagnostic result: the validation-FPR rule is mis-specified for this highly imbalanced project-held-out regime because

Table 11

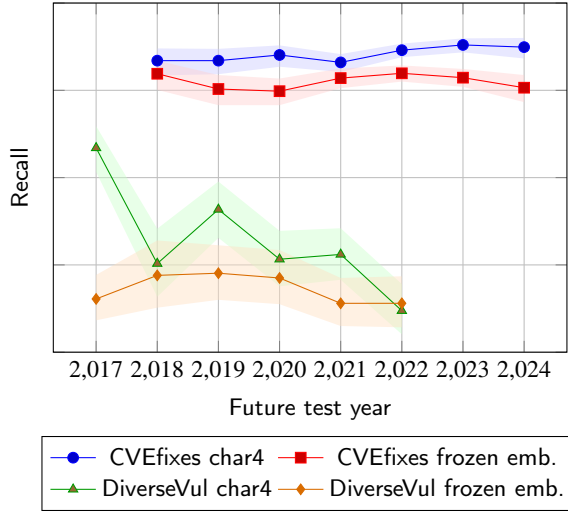
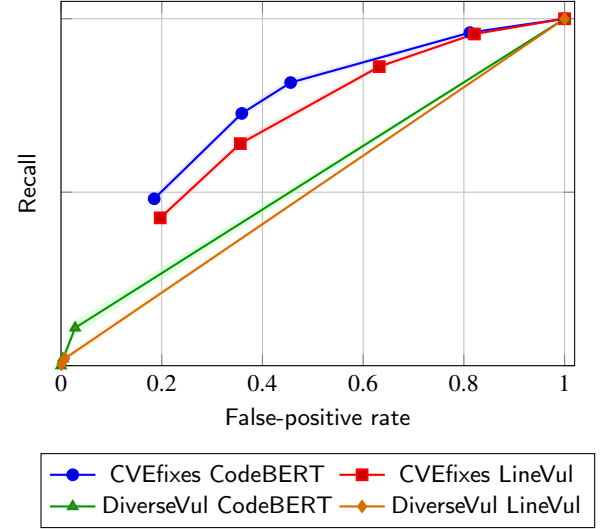
Near-duplicate-controlled temporal sensitivity and lightweight neural-transfer baselines.

Dataset	Model	First recall	Last recall	Mean recall	Mean F1
CVEfixes controlled	char4 hash LR	0.868	0.899	0.882	0.619
CVEfixes controlled	CodeBERT frozen emb. LR	0.827	0.853	0.810	0.643
CVEfixes controlled	LineVul calibrated	0.147	0.240	0.166	0.229
DiverseVul recovered	char4 hash LR	0.668	0.295	0.455	0.180

Table 12

Full-dataset neural fine-tuning under unseen-project temporal testing.

Dataset	Model	Train	Valid.	Test	Valid. recall	Valid. FPR	Mean recall	Mean F1
CVEfixes	CodeBERT full FT	3,980	579	11,369	0.615	0.190	0.563	0.612
CVEfixes	LineVul full FT	3,980	579	11,369	0.452	0.183	0.414	0.503
DiverseVul recovered	CodeBERT full FT	11,208	874	15,724	0.000	0.000	0.000	0.000
DiverseVul recovered	LineVul full FT	11,208	874	15,724	0.000	0.000	0.006	0.011

**Figure 5:** Fixed-origin future recall curves with uncertainty bands.**Figure 6:** Threshold/FPR tradeoff for project-held-out neural runs.

it can meet the FPR ceiling by selecting a near-all-negative threshold.

Table 13 is treated as a primary result rather than an appendix-style diagnostic: neural detector claims in this article are interpreted through operating-point rules, not through a single selected threshold. It expands the same neural runs across operating points, including best-F1, FPR-controlled, precision-floor, and Youden- J policies computed from the saved score distributions. On CVEfixes, CodeBERT has higher AUROC and AUPRC than LineVul, but both models trade recall against false-positive rate sharply as the threshold changes. At the best-F1 threshold, CodeBERT reaches recall 0.816 at FPR 0.456 and F1 0.695; under the FPR20 operating point, recall falls to 0.481 at FPR 0.185 and F1 to 0.567. The precision-floor policy recovers high recall on CVEfixes only by accepting high FPR (0.812 for CodeBERT and 0.821 for LineVul). Recovered-date DiverseVul is weaker under every threshold rule: CodeBERT's best-F1 and Youden- J points both reach only recall 0.109 at FPR 0.028 and F1 0.145, while the precision-floor rule is infeasible and falls back to precision 0.263 with recall 0.020 at FPR 0.004. Thus the earlier zero-recall point was a threshold artifact, but the rerun still shows no useful neural operating point on the recovered-date project-held-out split. In Table 13, FPR20 denotes validation-threshold selection with FPR at most 0.20, and PF50 denotes the best available precision-floor point at precision at least 0.50, falling back

to the highest-precision non-empty point when the floor is infeasible.

5.3. Independent Clean and Grouped Validity

Table 14 summarizes grouped and independent-clean checks. The near-duplicate-controlled runs have no train-to-future commit-group or paired vulnerable/fixed-group overlap. Project overlap is low in future CVEfixes test windows and moderate in recovered-date DiverseVul. The repository-clean benchmark gives the strongest warning: the CVEfixes-trained character model has specificity 0.155 on clean functions harvested from projects outside the vulnerability-dataset construction pipeline. This confirms that fixed-code negatives do not substitute for independently sampled clean functions.

Table 15 reports the repository-clean benchmark by detector family with Wilson confidence intervals. The CVEfixes-trained detectors have poor specificity across all four feature families, ranging from 0.075 to 0.275. Recovered-date DiverseVul is less uniformly poor: code metrics, frozen embeddings, and character features have specificity between 0.700 and 0.825, while token TF-IDF still flags many repository-clean functions. The key validity point remains: a model can look strong on pre/post-fix temporal recall while flagging many clean-only functions.

Table 13

Operating-point summary for full-dataset project-held-out neural runs.

Dataset	Model	AUROC	AUPRC	Best-F1 R/FPR	FPR20 R/FPR	PF50 P/R/FPR	Youden R/FPR
CVEfixes	CodeBERT full FT	0.737	0.682	0.816/0.456	0.481/0.185	0.504/0.960/0.812	0.727/0.359
CVEfixes	LineVul full FT	0.688	0.618	0.862/0.632	0.426/0.197	0.501/0.956/0.821	0.640/0.356
DiverseVul recovered	CodeBERT full FT	0.724	0.149	0.109/0.028	0.109/0.028	0.263/0.020/0.004	0.109/0.028
DiverseVul recovered	LineVul full FT	0.696	0.124	1.000/1.000	0.021/0.008	0.222/0.004/0.001	0.021/0.008

Table 14

Grouped validity and independent-clean specificity.

Dataset	Test project overlap	Commit overlap	Pair overlap	Unseen-project recall	Clean specificity
CVEfixes controlled	0.046	0.000	0.000	0.891	0.155
DiverseVul recovered	0.102	0.000	0.000	0.494	0.700

RQ1 summary. CVEfixes shows stable pre/post-fix recall for the primary character baseline, but the high temporal FPR and low repository-clean specificity limit any operational interpretation. Recovered-date DiverseVul is more volatile, showing that temporal behavior depends on dataset construction and operating point.

5.4. RQ2: Drift Sources

Rolling-origin drift analysis produces 48 paired drift-performance rows rather than relying on a single annual sequence. For CVEfixes, the strongest adjusted association is project turnover with recall drop across 26 paired replicates ($r = -0.558$, Fisher 95% CI $[-0.777, -0.218]$, $p_{\text{Holm}} = 0.027$). Vulnerable-ratio shift, CWE divergence, and project divergence are also moderate in magnitude, but only project turnover remains below 0.05 after Holm adjustment. For recovered-date DiverseVul, CWE turnover is largest across 22 paired replicates ($r = 0.422$, Fisher

95% CI $[0.001, 0.716]$), but it does not survive family-wise correction ($p_{\text{Holm}} = 0.445$). We therefore treat drift signals as maintenance indicators, not causal explanations: they identify windows that deserve inspection, but they do not by themselves prove why performance changed.

RQ2 summary. Rolling-origin analysis gives more evidence than a single annual sequence: project turnover is the strongest adjusted CVEfixes signal, whereas CWE turnover is only descriptive for recovered-date DiverseVul after correction. Drift signals are therefore maintenance indicators, not causal explanations.

5.5. RQ3: Maintenance Policies

Table 19 reports the ordinary-temporal maintenance experiment, whereas Table 8 reports the stricter project-held-out version. The recall values therefore differ by design. On CVEfixes, all ordinary-temporal refresh policies improve recall over no refresh, but they also operate at high false-positive rates. The small recent update policy reaches mean

Table 15

Repository-clean specificity by detector family.

Training source	Model	Rows	Specificity	95% CI	FPR
CVEfixes	char4 hash LR	2500	0.155	$[0.142, 0.170]$	0.845
CVEfixes	code metrics LR	2500	0.075	$[0.065, 0.086]$	0.925
CVEfixes	frozen emb. LR	2500	0.275	$[0.258, 0.293]$	0.725
CVEfixes	token TF-IDF LR	2500	0.184	$[0.170, 0.200]$	0.816
DiverseVul recovered	char4 hash LR	2500	0.700	$[0.682, 0.718]$	0.300
DiverseVul recovered	code metrics LR	2500	0.825	$[0.810, 0.840]$	0.175
DiverseVul recovered	frozen emb. LR	2500	0.812	$[0.796, 0.826]$	0.188
DiverseVul recovered	token TF-IDF LR	2500	0.579	$[0.560, 0.598]$	0.421

Table 16

Repository-clean specificity by language.

Training source	Clean language group	Rows	Specificity	95% CI	Clean FPR
CVEfixes	C/C++	2265	0.159	$[0.144, 0.175]$	0.841
CVEfixes	Python	235	0.119	$[0.084, 0.167]$	0.881
DiverseVul recovered	C/C++	2265	0.694	$[0.674, 0.712]$	0.306
DiverseVul recovered	Python	235	0.766	$[0.708, 0.816]$	0.234

Table 17

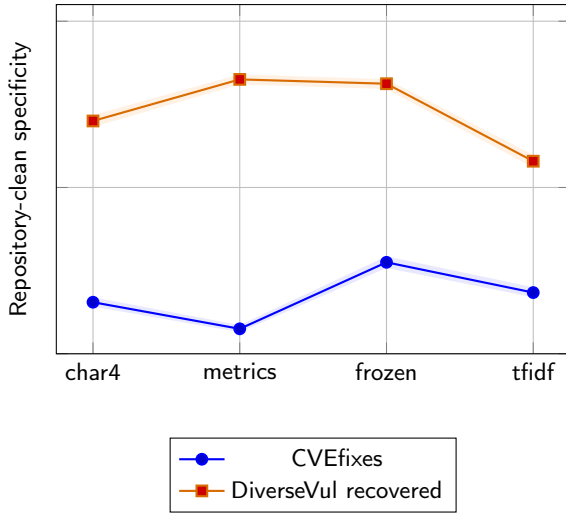
Rolling-origin paired drift–performance analysis.

Dataset	Strongest adjusted signal	Pairs	Pearson r	95% CI	Holm p
CVEfixes	Project turnover	26	-0.558	[-0.777, -0.218]	0.027
DiverseVul recovered	CWE turnover	22	0.422	[0.001, 0.716]	0.445

Table 18

Compact statistical-output summary for the primary character baseline.

Dataset	Effect	Estimate	Interval or adjusted p	Cost	Interpretation
CVEfixes	Ordinary temporal recall change	+0.036	–	–	No evidence of monotonic aging for the primary baseline.
CVEfixes	Small-update recall gain vs no refresh	+0.072	[0.934, 0.974]	40 labels/window	Low-cost update improves recall in ordinary temporal testing.
CVEfixes	Project-held-out small-update FPR	0.945	–	40 labels/window	Recall gain is useful only at an explicitly accepted FPR operating point.
CVEfixes	Project turnover vs recall drop	$r = -0.558$	CI [-0.777, -0.218]; $p_H = 0.027$	$n = 26$	Rolling-origin association after Holm correction.
DiverseVul recovered	Ordinary temporal recall change	-0.372	–	–	Recovered-date subset shows stronger temporal decline.
DiverseVul recovered	Project-held-out no-refresh FPR	0.860	–	–	High recall is coupled to high false-positive cost.
DiverseVul recovered	C2 service-level check	no pass	recall ≥ 0.50 and FPR ≤ 0.30	six policies	Negative result: maintenance alone does not reach the specified operating point.
DiverseVul recovered	CWE turnover vs recall drop	$r = 0.422$	CI [0.001, 0.716]; $p_H = 0.445$	$n = 22$	Descriptive only after rolling-origin correction.

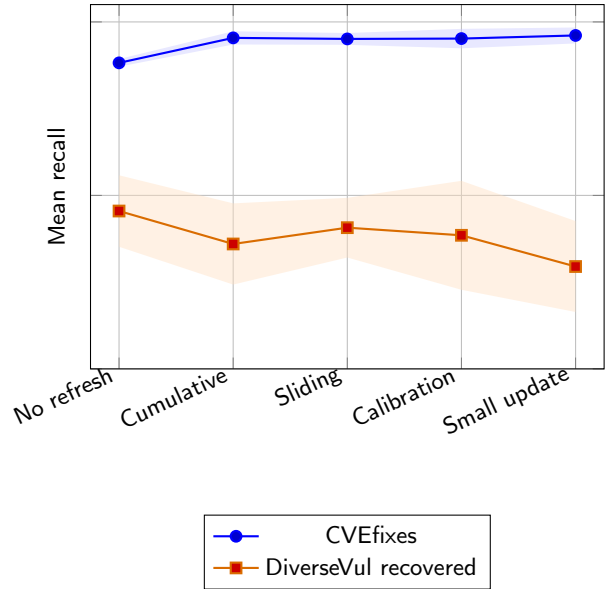
**Figure 7:** Repository-clean specificity benchmark.

recall 0.961 at mean FPR 0.928 while using 40 recent labels per future window. On recovered-date DiverseVul, fixed-origin evaluation has higher recall than retraining variants, whereas small recent update has lower recall 0.295 but lower FPR 0.137. This negative result is important for the maintenance framing: update policies are interventions whose benefits depend on the future distribution and the operating objective.

RQ3 summary. Maintenance policies change recall–FPR trade-offs rather than simply improving detectors. Small recent updates improve CVEfixes recall at high FPR, while no recovered-date DiverseVul policy reaches the C2 service level of recall ≥ 0.50 at FPR ≤ 0.30 .

5.6. RQ4: Exploratory Monitoring Calibration

Monitoring triggers are evaluated against a pre-specified recall-floor or recall-drop event. In the CVEfixes primary run, no yearly window violates the chosen event definition for the character baseline, so trigger precision and recall are structurally uninformative under that service level. In the recovered-date DiverseVul run, CWE divergence, project

**Figure 8:** Maintenance-policy mean recall with bootstrap uncertainty.

turnover, and unseen-token rate each achieve precision 1.000 and recall 0.750 for the event definition, but this corresponds to only three detected event windows out of four. RQ4 is therefore demoted to an exploratory monitoring-calibration check. It does not support a mature trigger-learning claim; that would require rolling-origin replicates or another dataset to raise the number of service-level events from single digits to dozens. The limited result is still useful as a protocol guardrail: drift alarms should be interpreted only after an operational service level and maintenance action have been specified.

RQ4 summary. The available yearly windows do not support a mature trigger-learning claim. CVEfixes has no violations under the chosen service event, and recovered-date DiverseVul yields only a small 3-of-4 event pattern, so drift alarms should be interpreted only with a pre-declared service level and action.

Table 19

Maintenance-policy summary for the selected primary baseline.

Dataset	Policy	Mean recall	Mean F1	Mean FPR	Recall CI
CVEfixes	No refresh	0.882	0.619	0.821	[0.871, 0.893]
CVEfixes	Cumulative retrain	0.954	0.630	0.912	[0.935, 0.971]
CVEfixes	Sliding 3-year retrain	0.951	0.629	0.908	[0.933, 0.967]
CVEfixes	Calibration only	0.952	0.627	0.916	[0.924, 0.969]
CVEfixes	Small recent update	0.961	0.629	0.928	[0.938, 0.978]
DiverseVul recovered	No refresh	0.455	0.180	0.252	[0.373, 0.558]
DiverseVul recovered	Cumulative retrain	0.360	0.192	0.168	[0.265, 0.477]
DiverseVul recovered	Sliding 3-year retrain	0.407	0.188	0.201	[0.321, 0.489]
DiverseVul recovered	Calibration only	0.385	0.171	0.211	[0.229, 0.542]
DiverseVul recovered	Small recent update	0.295	0.180	0.137	[0.208, 0.426]

6. Discussion

6.1. What the Results Say

The strongest result is not that every detector inevitably ages. The strongest result is that vulnerability-detector maintenance policies and operating points materially affect future temporal behavior under project and label-distribution shift. The same problem can look stable, degraded, or unusable depending on temporal cutoff, feature family, threshold policy, project holdout, and whether independently clean negatives are included. These are exactly the kinds of choices software maintenance research is designed to make explicit.

6.2. Practical Implications

Teams deploying learning-based vulnerability detectors should:

- evaluate on chronological splits and, when possible, unseen projects;
- report false-positive operating points rather than only maximum F1;
- keep independent clean functions for specificity testing;
- track drift in project mix, CWE mix, vocabulary, code length, and class prior;
- compare calibration, retraining, and small recent updates as explicit maintenance policies;
- define service-level triggers before interpreting drift alarms.

Table 20 translates these implications into concrete maintenance choices.

Table 21 gives the same point in operational units. It is not a universal cost model; it simply converts FPR into expected false alerts for a chosen review surface. Teams should replace the example surfaces with their own pull-request or CI volumes before selecting a service-level objective.

7. Threats to Validity

Construct validity of labels. CVEfixes compares vulnerable and fixed versions of changed code. This is a meaningful pre/post-fix task, but it is not the same as arbitrary vulnerable-versus-clean detection. Claims based on CVEfixes must therefore be read as claims about pre/post-fix detection unless they are supported by independent clean examples. The independent-clean challenge shows that CVEfixes-trained models can retain high temporal recall while having poor external clean specificity.

Temporal metadata. CVEfixes has explicit temporal metadata in the local mirror. DiverseVul requires enrichment from external repository metadata. The recovery pass substantially improves coverage, but 10,478 local rows still lack recovered dates (Table 3). We therefore use recovered-date DiverseVul as sensitivity evidence and exclude unrecovered rows from temporal claims rather than mixing proxy and recovered dates. This choice sacrifices coverage to avoid a more damaging validity problem: treating heterogeneous date sources as if they were equivalent.

Commit-date external replication. PrimeVul-paired adds a third dataset with recovered commit dates rather than CVE-year proxy dates. Coverage is partial: 494 commit pages are recovered, yielding 1,267 post-deduplication rows from 74 projects. Its results are therefore used as external sensitivity evidence, not as a replacement primary dataset. The paired construction also means the task remains pre/post-fix detection, not arbitrary vulnerable-versus-clean detection.

Detector-family validity. The study includes lightweight baselines, frozen CodeBERT, calibrated LineVul, and full-dataset fine-tuning. Bounded fine-tuning is retained only as an artifact-level execution check because its sample size, one-epoch schedule, and maximum length 128 are too limited for architecture comparison. The recovered-date DiverseVul neural collapse is primarily a threshold-policy artifact: the fixed-FPR validation rule selects a near-all-negative threshold even though the ranking metrics are non-random. The PrimeVul-paired

Table 20

Practical maintenance-policy recommendations.

Deployment condition	Recommended action	Rationale
Recall is the primary service objective	Compare small recent update and calibration-only policies.	These policies improve CVEfixes recall, but must still be checked against false-positive cost.
False positives are constrained	Select thresholds with validation FPR caps and report operating-point curves.	Best-F1 thresholds can hide high alert rates; FPR-controlled thresholds expose the deployment trade-off.
Future projects differ from training projects	Use project-held-out validation and testing.	Ordinary chronological splits do not measure generalization to unseen projects.
Training negatives are fixed code versions	Require independent clean functions and report specificity by source.	Fixed-code negatives do not establish arbitrary vulnerable-versus-clean behavior.
Drift monitors trigger an alarm	Link the alarm to a pre-declared service event and maintenance action.	Drift alone is not failure; it becomes useful only when tied to recall, FPR, or label-cost objectives.

Table 21

Illustrative alert-cost conversion.

Operating condition	Clean funcs	FPR	Expected false alerts
Strict validation cap	20 per PR	0.200	4.0; high but triageable
Cross-dataset stress	20 per PR	0.700	14.0; likely too noisy
CVEfixes project-held-out small update	20 per PR	0.945	18.9; nearly every clean function flagged
Repository-clean char4 check	1,000 in CI	0.845	845; clean benchmark failure

recovered-date neural rerun addresses part of this concern by retaining training curves, validation losses, validation AUPRC/AUROC, and validation threshold curves under five requested epochs, length-256, class-balanced fine-tuning on all eligible recovered-date rows. It remains diagnostic, however, not a definitive neural architecture comparison. A stronger neural operating-point claim would require a larger learning-rate sweep, external prospective validation, and additional detector families such as CodeT5, UniXcoder, graph-based vulnerability detectors, and multi-modal vulnerability models. The contribution is therefore a maintenance-evaluation framework and empirical stress test, not a claim that the evaluated neural architectures exhaust the state of the art.

Leakage controls. Exact duplicates are removed before splitting, and near-duplicate-controlled reruns remove future records close to earlier windows. Approximate matching thresholds may still under-match semantic clones or over-match short boilerplate functions.

Project and group validity. The grouped audit finds no train-to-future commit-group or vulnerable/fixed-pair overlap in controlled runs, and the full-dataset neural rerun reports unseen-project temporal evaluation. The classical maintenance-policy rerun applies the same unseen-project temporal constraint, although its logistic models are bounded and sampled for tractable reruns. These project-held-out classical policies reveal very high FPR on CVEfixes, so their recall gains should be interpreted as maintenance tradeoffs, not improvements at an acceptable operating point by themselves.

Statistical validity. Annual temporal windows are few, so correlations between drift and recall should be read as descriptive associations. We report the number of windows, confidence intervals for correlations, permutation tests, and Holm-adjusted values over the drift-signal family, but these tests still do not establish causality.

8. Conclusion

This article introduced MAINT-VD, a maintenance-evaluation framework for studying learning-based vulnerability detectors as maintained software artifacts. A deployed detector has a lifecycle: it is trained on historical projects, calibrated on a validation set, operated at a chosen threshold, monitored against future code, and periodically refreshed or revalidated as the surrounding software ecosystem changes. Treating these decisions as part of the detector turns temporal evaluation into a software maintenance problem.

The empirical results support a precise conclusion: maintenance policies and threshold rules are part of detector behavior under project and label-distribution shift; temporal recall is meaningful only when reported with false-positive and clean-specificity evidence. The evidence is not a simple story that every detector ages monotonically. On CVEfixes, the primary character baseline remains relatively stable in ordinary temporal evaluation, and low-cost recent updates improve already strong recall. Under project-held-out temporal testing, however, classical refresh policies obtain high recall at very high false-positive rates. Full-dataset CodeBERT and LineVul runs show the same

lesson from a neural perspective: AUROC and AUPRC summarize ranking behavior, but maintenance decisions require precision, recall, F1, specificity, and FPR at explicit operating points. PrimeVul-paired strengthens the external-validity story without changing this conclusion: after commit-date recovery, it provides a smaller but true-dated paired sensitivity study. Classical models still trade recall against high FPR, and the fuller neural rerun shows that retained training curves and more complete local training do not by themselves create a reliable future-project operating point.

The study also exposes two validity lessons for future vulnerability-detector research. First, fixed-code negatives are not sufficient evidence of clean-code specificity. The independent-clean challenge shows that a model can preserve future recall on a pre/post-fix task while flagging many independently clean functions. Second, temporal evidence depends on metadata quality. CVEfixes can serve as the primary temporal study because explicit dates are available in the local mirror. DiverseVul becomes much more useful after repository-date recovery, but remaining missingness still makes it better suited to sensitivity evidence than to the article's primary claim. These points sharpen rather than weaken the maintenance thesis: they specify the empirical conditions under which the thesis can be defended.

For practitioners, the implication is operational. A vulnerability detector should be deployed with a declared service level, a threshold-selection rule, an independent clean evaluation set, drift monitors, and a planned policy for calibration or retraining. For researchers, the implication is methodological: MAINT-VD is a reusable evaluation scaffold, not a final ranking of detector architectures. Temporal splits, project-held-out evaluation, near-duplicate controls, operating-point curves, and label-cost reporting should accompany claims about security-detector robustness. A single random split or a single F1-optimized threshold is not enough to describe how a detector will behave as software projects evolve.

Future work should recover repository commit dates for additional datasets, including Big-Vul and language-specific corpora such as Vul4J [21], extend the clean-function challenge across more languages and project sources, evaluate larger neural tuning budgets, and test the monitoring triggers in prospective or continuously updated settings. The broader goal is to make learning-based security tools auditable over time. Security detectors will be most useful when their maintenance assumptions are visible, measurable, and revisable like the assumptions of other software engineering infrastructure.

CRedit authorship contribution statement

Phat T. Tran-Truong: Conceptualization, Methodology, Software, Validation, Investigation, Writing - original draft, Writing - review and editing, Visualization. **Xuan-Bach Le:** Methodology, Validation, Writing - review and editing.

Declaration of Competing Interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Code Availability

The source code, experiment scripts, manuscript sources, and replication materials are available in the GitHub repository <https://github.com/LeoTTTPhat/MAINT-VD>. The repository should be archived with a DOI or immutable release tag before final submission so that the released artifact version matches the checksums reported in the repository manifest.

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References

- [1] Y. Zhou, S. Liu, J. Siow, X. Du, Y. Liu, Devign: Effective vulnerability identification by learning comprehensive program semantics via graph neural networks, in: *Advances in Neural Information Processing Systems*, 2019, pp. 10197–10207.
- [2] M. Fu, C. Tantithamthavorn, LineVul: A transformer-based line-level vulnerability prediction, in: *Proceedings of the International Conference on Mining Software Repositories*, 2022.
- [3] Z. Feng, D. Guo, D. Tang, N. Duan, X. Feng, M. Gong, L. Shou, B. Qin, T. Liu, D. Jiang, M. Zhou, CodeBERT: A pre-trained model for programming and natural languages, in: *Findings of the Association for Computational Linguistics: EMNLP*, 2020.
- [4] J. Fan, Y. Li, S. Wang, T. N. Nguyen, A c/c++ code vulnerability dataset with code changes and cve summaries, in: *Proceedings of the International Conference on Mining Software Repositories*, 2020.
- [5] G. P. Bhandari, A. Naseer, L. Moonen, CVEfixes: Automated collection of vulnerabilities and their fixes from open-source software, in: *Proceedings of the 17th International Conference on Predictive Models and Data Analytics in Software Engineering*, 2021, pp. 30–39. doi:10.1145/3475960.3475985.
- [6] Y. Chen, Z. Ding, L. Alowain, X. Chen, D. Wagner, DiverseVul: A new vulnerable source code dataset for deep learning based vulnerability detection, in: *Proceedings of the International Symposium on Research in Attacks, Intrusions and Defenses*, 2023.

- [7] D. Sculley, G. Holt, D. Golovin, E. Davydov, T. Phillips, D. Ebner, V. Chaudhary, M. Young, J.-F. Crespo, D. Dennison, Hidden technical debt in machine learning systems, in: *Advances in Neural Information Processing Systems*, Vol. 28, 2015.
- [8] S. Amershi, A. Begel, C. Bird, R. DeLine, H. Gall, E. Kamar, N. Nagappan, B. Nushi, T. Zimmermann, Software engineering for machine learning: A case study, in: *Proceedings of the IEEE/ACM 41st International Conference on Software Engineering: Software Engineering in Practice*, 2019, pp. 291–300. doi:10.1109/ICSE-SEIP.2019.00042.
- [9] M. Jimenez, M. Papadakis, Y. Le Traon, Vulnerability prediction models: A case study on the linux kernel, in: *Proceedings of the IEEE 16th International Working Conference on Source Code Analysis and Manipulation*, 2016, pp. 1–10. doi:10.1109/SCAM.2016.15.
- [10] R. Croft, M. A. Babar, M. Kholoosi, Data quality for software vulnerability datasets, in: *Proceedings of the IEEE/ACM 45th International Conference on Software Engineering*, 2023.
- [11] S. Chakraborty, R. Krishna, Y. Ding, B. Ray, Deep learning based vulnerability detection: Are we there yet?, *IEEE Transactions on Software Engineering* (2022).
- [12] D. Q. Nguyen, K. Rieck, L. Grunske, C. Kästner, Revisiting graph neural networks for vulnerability detection, in: *Proceedings of the International Conference on Software Engineering*, 2022.
- [13] B. Steenhoek, M. M. Rahman, R. Jiles, W. Le, An empirical study of deep learning models for vulnerability detection, in: *Proceedings of the IEEE/ACM 45th International Conference on Software Engineering*, 2023, pp. 2237–2248. doi:10.1109/ICSE48619.2023.00188.
- [14] N. Risse, M. Böhme, Uncovering the limits of machine learning for automatic vulnerability detection, in: *Proceedings of the USENIX Security Symposium*, 2024.
- [15] Y. Wang, W. Wang, S. Joty, S. C. H. Hoi, CodeT5: Identifier-aware unified pre-trained encoder-decoder models for code understanding and generation, in: *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, 2021, pp. 8696–8708.
- [16] D. Guo, S. Lu, N. Duan, Y. Wang, M. Zhou, J. Yin, UniXcoder: Unified cross-modal pre-training for code representation, in: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, 2022, pp. 7212–7225.
- [17] Y. Ding, et al., PrimeVul: A comprehensive dataset for function-level vulnerability detection, *arXiv preprint arXiv:2403.18624* (2024).
- [18] Z. Li, et al., ReposVul: A repository-level high-quality vulnerability dataset, *arXiv preprint arXiv:2401.13169* (2024).
- [19] W. Lu, et al., ICVul: A well-labeled c/c++ vulnerability dataset with comprehensive metadata and vulnerability-contributing commits, *arXiv preprint arXiv:2505.08503* (2025).
- [20] J. Gama, I. Zliobaite, A. Bifet, M. Pechenizkiy, A. Bouchachia, A survey on concept drift adaptation, *ACM Computing Surveys* 46 (4) (2014) 1–37. doi:10.1145/2523813.
- [21] Q.-C. Bui, R. Scandariato, N. E. D. Ferreyra, Vul4J: A dataset of reproducible java vulnerabilities geared towards the study of program repair techniques, in: *Proceedings of the IEEE/ACM 19th International Conference on Mining Software Repositories*, 2022, pp. 464–468. doi:10.1145/3524842.3528482.